KLH UNIVERSITY HYDERABAD

PROJECT PRESENTATION

Department of Computer Science and Engineering

KLH Bachupally Campus, Hyderabad

Team Number T.N. – 18

Guide Name Mr. Shaik Asif

Project Title : Customer Churn Prediction Using Elastic Net Regularized Logistic Regression

INTRODUCTION

- Predicts customers likely to leave a telecom service
- Uses the Telco Customer Churn dataset
- Combines Logistic Regression with Elastic Net regularization
- Helps identify at-risk customers for proactive retention



RESEARCH GAP

- Existing models may overfit and fail to generalize
- Customer features are often correlated
- Limited interpretability in black-box models
- Need for a balanced and explainable solution

CURRENT RESEARCH	OUR SOLUTION
• Unregularized models • Overfitting risk • High feature correlation	• Elastic Net regularization • Better generalization • Balance of prediction and interpretability

ALGORITHM



GRAPHS & RESULTS

ROC-AUC	Recall	F1-Score	Accuracy
84.8%	55.2%	60.1%	80.5%
Ranking discriminalion	Churn cases identified	Precision / recall balance	Correct classifications

Churn by contract

Percentage of customers who churned

Month-to-month	42.7%
One year	11.3%
Two	2.8%

Month-to-month contracts show the highest historical churn rate in this dataset.

Churn by internet service

Historical churn rate by service type

DSL	19.0%
Fiber optic	78.3%
No	7.4%

Service category is associated with different churn rates; this does not establish causation.

Innovation | Research | Impact

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RESULTS – FRONT END

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1. Login Page



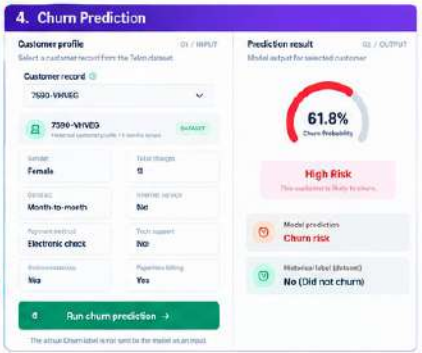
2. Dashboard (Home Page)



3. Customer Records



4. Churn Prediction



5. Reports & Analytics



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